

TECHNICAL DOCUMENTATION · v1

Gaze Detection

Webcam Eye-Tracking System



Full engineering reference: architecture, feature set, distance model, calibration mathematics, filtering, and the game layer. Real-time gaze estimation from a standard webcam — no infrared hardware.

~30

FPS end-to-end (GPU)

478

facial landmarks

degree-3

calibration polynomial

32

input features

STACK

Python · OpenCV · MediaPipe FaceLandmarker · L2CS-Net (ResNet-50, Gaze360) · PyTorch + CUDA · scikit-learn (ridge) · One Euro filter · pytest

1 System Overview

The system converts a live webcam video of a face into a continuous on-screen gaze coordinate, plus a confidence value and an interaction state (moving / dwell). It runs in real time on consumer hardware with no infrared emitters or head-mounted equipment — the constraint that defines the project.



Design principle

architecture

- MODULAR** Each stage is independent and unit-tested with synthetic inputs — the maths is verified without a camera.
- FAIL-SAFE** On a blink, lost face, or covered eyes, the gate freezes output rather than feeding corrupted values downstream.
- PER-USER** A generic gaze model can't know your screen or seating; a fast calibration adapts it to each person and session.

2 The Feature Vector

Every frame is described by a compact set of numbers derived from the 478 landmarks. These feed the calibration model. They fall into three families: eye geometry (where the eyes point), head pose (where the head points), and position (where the head is).

FEATURE	FAMILY	WHAT IT ENCODES
<code>gaze_pitch / gaze_yaw</code>	Eye	Eyeball direction up/down and left/right, from L2CS-Net.
<code>iris_offset (per eye)</code>	Eye	Iris position inside the eye opening, scaled to [-1,1].
<code>head_pitch / yaw / roll</code>	Head	Head orientation — from MediaPipe's 4×4 transform matrix.
<code>pos_distance_cm</code>	Position	Distance from the screen in cm, from inter-pupil pixels.
<code>face_centre_offset</code>	Position	Head translation left/right and up/down in the frame.

FEATURE	FAMILY	WHAT IT ENCODES
inter-ocular px	Position	Apparent eye separation — the raw distance signal.

Head pose vs eye gaze are separate on purpose. You can hold your head still and move only your eyes, or stare straight ahead while turning your head. The model needs both signals to disentangle where the eyes truly land on screen.

3 The Distance Model

Eye angle alone is ambiguous — the same angle lands on a different screen point depending on how far you sit. The system recovers absolute distance from a single known constant: adult inter-pupillary distance is ~6.3 cm.

```
distance_cm = focal_px × 6.3 / ipd_px
```

Pinhole-camera distance

`pos_distance_cm`

`ipd_px` Pixels between the two irises (landmarks 468 ↔ 473).

`focal_px` Camera focal length in pixels — 1073 on the dev laptop, found via a one-time 'press c at 50 cm' calibration.

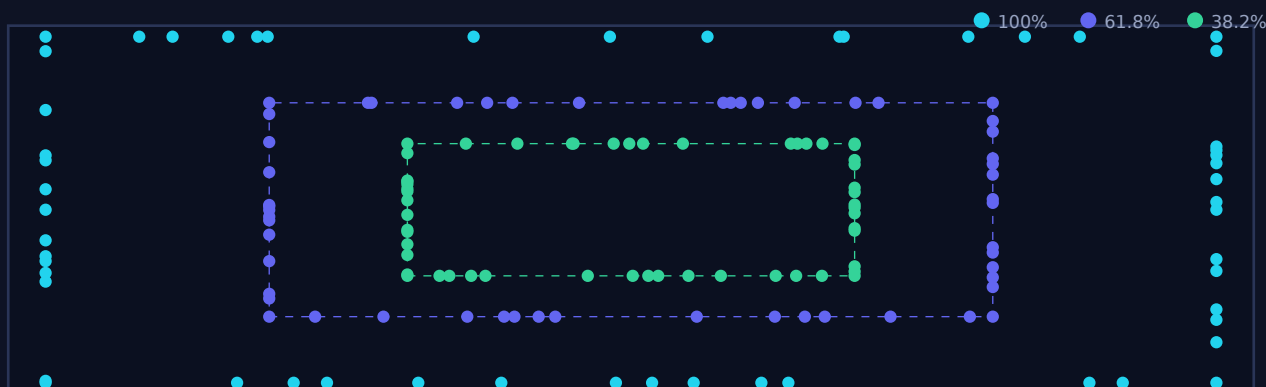
EXAMPLE $1073 \times 6.3 / 130 \text{ px} = 52 \text{ cm}$.

GATE The positioning gate keeps the user inside the calibrated 45–65 cm band; outside it, the user is prompted to adjust.

Analogy — a camera-only reversing car. With no ultrasonic sensor, it judges distance by knowing a number plate's real width, measuring how wide it appears, and inverting. Here the “number plate” is the 6.3 cm pupil gap.

4 Calibration Mathematics

Calibration learns the per-user map from feature vector → screen position. Targets are placed as polygon vertices at multiple scales for full-screen coverage; the map is a regularised polynomial fit.



Vertices of line → decagon on the screen rectangle (square pinned to the corners), repeated at 100 / 61.8 / 38.2 % scale. Edges, corners and interior all receive targets — 162 total.

The fitted map

calibration_model.pkl

MODEL	A degree-3 polynomial with 10 terms mapping features → (x, y) screen position.
POSTURE	Collection is split into posture blocks (shift, lean, tilt) so the map learns head-movement compensation — otherwise it breaks the moment you move live.
CENTRE OFFSET	A start-of-session centre dot measures each user's constant gaze offset (kappa angle), subtracted from everything after.

Corner coverage matters. A plain left-right sweep touches the edges only briefly, so the fit had to extrapolate at the corners — top-right was off by **6.2 cm**. A slow perimeter lap added corner data and cut it to **0.7 cm**.

5 Smoothing & Interaction

Raw predictions are noisy. The One Euro filter suppresses jitter adaptively, and a dwell detector turns steady gaze into a selectable “eye-click.”

One Euro filter

adaptive low-pass

- BEHAVIOUR** Cutoff frequency scales with gaze speed: still gaze smooths hard, fast gaze follows instantly.
- RESULT** Measured jitter dropped from ~114 px to ~74 px with no perceptible lag.
- VS EMA** A fixed average forces one lag/smoothness trade-off; One Euro adapts per frame.

Dwell detection

eye-click

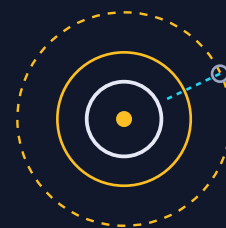
- RULE** Gaze staying within a small radius for a set time fires a dwell event.
- USE** The interaction primitive for hands-free selection — buttons, tiles, menus.
- FEEDBACK** An expanding ring animation confirms the click at the dwell point.

6 The Game Layer

A gaze game sits on the same pipeline. It serves two purposes beyond play: aim assist makes small targets hittable at ~2 cm accuracy, and every round logs labelled gaze-vs-target data for future model improvement.

Aim assist

- PULL** $\text{assisted} = \text{raw} + (\text{target} - \text{raw}) \times \text{pull}$; pull grows 0 → 1 from field edge to snap zone.
- WHY** At ~2 cm accuracy, small targets are hard to hit even when looked at directly.
- DATA** Each hit/miss logs looked-at vs true centre — labelled data for retraining.



raw gaze pulled toward target

ZONE	SIZE	VISIBL E	BEHAVIOUR
Target ring	70 px / 1.9 cm	Yes	Dwell 0.8 s inside → target pops.
Snap zone	~140 px	No	Cursor locks onto target centre.

ZONE	SIZE	VISIBL E	BEHAVIOUR
Magnet field	~260 px	No	Cursor pulled gradually inward.

Validation metrics

how we measure

METHOD	A ball follows a path; recorded gaze is compared against it (per Vidhya & Faria 2025).
METRICS	MAD, RMSE, and accuracy = % of gaze points within a threshold radius of the ball.
SEPARATION	Bias (accuracy, from calibration) and jitter (steadiness, from smoothing) are reported separately.